

Project Name: Automated Network Request Management System

1. INTRODUCTION

1.1 Project Overview

The **Automated Network Request Management System** is a ServiceNow-based solution that simplifies network provisioning and office relocation requests. It replaces manual email-based processes with a centralized self-service portal, enabling employees to submit requests efficiently while reducing delays and improving data accuracy.

1.2 Purpose

The project aims to automate the network request process by minimizing manual effort, reducing processing time, and ensuring accurate data collection. It improves transparency through automated approvals and real-time request tracking while maintaining secure and reliable records.

2. IDEATION PHASE

2.1 Problem Statement

The existing network request process relies on emails and manual documentation, leading to incomplete requests, approval delays, limited visibility, and data entry errors. These issues reduce productivity and delay workspace readiness during office relocations.

2.2 Empathy Map Canvas

User Says: "I don't know the status of my network request."

User Thinks: "The process should be faster and more transparent."

User Does: Frequently follows up with the IT team for updates.

User Feels: Frustrated due to delays and lack of communication.

2.3 Brainstorming

The team identified three key solution areas:

- A centralized Service Portal for request submission.
- Automated approval workflows using Flow Designer.
- A dedicated custom database (**u_network_database**) for secure data storage.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Employee accesses the Service Portal.
2. User details are auto-filled and the form adapts based on request type.
3. The request is routed for manager approval with status notifications.
4. Approved requests are stored in the database and confirmation is displayed.

3.2 Solution Requirements

Functional Requirements

- Auto-populate employee details.
- Mandatory field validation.
- Dynamic UI policies.
- Automated approval workflow.
- Secure database storage.

Non-Functional Requirements

- Fast system performance.
- Secure role-based access.
- Reliable and scalable architecture.

3.3 Data Flow Diagram

Service Portal



Client Scripts & Validation



Flow Designer Approval



Manager Approval



u_network_database

3.4 Technology Stack

- **Frontend:** ServiceNow Service Portal
- **Workflow Automation:** Flow Designer
- **Backend:** Custom Table (u_network_database)

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The proposed solution addresses manual errors through validation, improves visibility with status tracking, and accelerates approvals using automated workflows.

4.2 Proposed Solution

A self-service portal integrated with Flow Designer automates request submission, approvals, notifications, and secure data storage.

4.3 Solution Architecture

The solution consists of four stages:

1. Portal Submission
 2. Workflow Processing
 3. Manager Approval
 4. Secure Database Update
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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project followed an Agile approach with two development sprints.

Sprint 1

- Auto-populate user details.
- Dynamic UI policies.
- Frontend validation.

Sprint 2

- Manager approval workflow.
- Database integration.
- Dashboard development.

Project Summary

- Total Story Points: **19**
 - Total Sprints: **2**
 - Average Velocity: **9.5 Story Points per Sprint**
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6. FUNCTIONAL & PERFORMANCE TESTING

6.1 Testing Summary

The application was tested to validate functionality, security, and performance.

Defect Resolution

- Critical defects fixed: **8**
- Minor UI defects fixed: **27**

Test Results

Module	Test Cases	Status
Service Portal	22	Passed
Flow Designer	18	Passed
ACL Security	12	Passed
Database Validation	15	Passed
Exception Handling	9	Passed

Total: 76 Test Cases Executed – 100% Pass Rate

7. RESULTS

The system successfully automated the complete network request lifecycle. Users can submit requests through the Service Portal, receive real-time updates, obtain automated approvals, and have approved records securely stored in the custom database.

8. ADVANTAGES & LIMITATIONS

Advantages

- Reduces manual effort and processing time.
- Improves data accuracy through validation.
- Provides real-time request tracking.
- Enhances security with role-based access.
- Increases overall operational efficiency.

Limitations

- Dependent on ServiceNow platform availability.
- Initial configuration and workflow setup require development effort.

9. CONCLUSION

The Automated Network Request Management System successfully streamlines enterprise network provisioning by replacing manual processes with an automated ServiceNow solution. It improves efficiency, ensures accurate data management, accelerates approvals, and provides a transparent experience for both employees and administrators.

10. FUTURE SCOPE

Future enhancements include AI-based request prioritization, predictive resource allocation, intelligent ticket classification, chatbot integration, and advanced analytics dashboards to further improve automation and user experience.

